

COMPUTATIONAL GENOMICS FINAL PROJECT

Who left this DNA?

A mystery Nanopore fecal sample, six possible host animals, one microbial co-star, and a small pile of evidence.

Group D

Host origin, microbiome, assembly, and validation from curated project outputs.

THE BRIEF

What the assignment actually asked us to do

Technical part

Clean and summarize the Nanopore reads. Explain why the data are usable.

Bioinformatics part

Find the host animal, classify microbes, assemble one abundant species, and validate it.

Biological part

Turn the outputs into a story: what animal, what microbe, what functional region, and how confident we are.

6

candidate host genomes

300k

reads used for the first host screen

1

dominant microbial species

1

reproducible notebook pipeline

COURSE CONTEXT

What we can assume from previous classes

Lesson 3: sequencing files and QC

FASTQ, read length, Phred quality, FastQC-style reports, and deciding whether data look usable.

Lesson 4: mapping and BAMs

Reference genomes, aligners, SAM/BAM, CIGAR, and MAPQ as a confidence score.

Lesson 5: visual inspection

IGV, coverage at a genomic coordinate, variant evidence, and long-read technologies including ONT.

Lesson 9: assembly

Reads overlap, k-mers, De Bruijn graphs, contigs, coverage, repeats, and why assemblies fragment.

So we do not re-teach

FASTQ basics, what alignment means, how MAPQ is defined, or why genome browsers are useful.

We do explain

How those ideas change for a fecal Nanopore microbiome sample.

ANALYSIS MAP

From messy reads to a biological story

1

Check the reads

Are they long enough and clean enough to trust?

2

Find the host

Align reads to the six candidate genomes.

3

Pick a region

Use gene annotations to find a readable functional-region example.

4

Classify microbes

Ask what else appears in the fecal DNA.

5

Assemble and validate

Reconstruct the dominant microbe and compare it to a reference.

The goal was not to make a fancy tool zoo. It was to make each step answer the next practical question.

WHAT IS NEW HERE?

Same course workflow, final-project version

FastQC idea, Nanopore tools

Instead of Illumina FastQC, we used NanoPlot/NanoComp because read length and quality behave differently for ONT reads.

BWA idea, minimap2 tool

Same mapping logic as Lesson 4, but minimap2 is better suited for long noisy reads.

IGV idea, host region check

Same visual sanity-check mindset as Lesson 5, now applied to a cat reference region.

New piece: microbiome classification

Kraken2 asks "which organisms do these reads resemble?" using a k-mer database.

Assembly idea, Flye tool

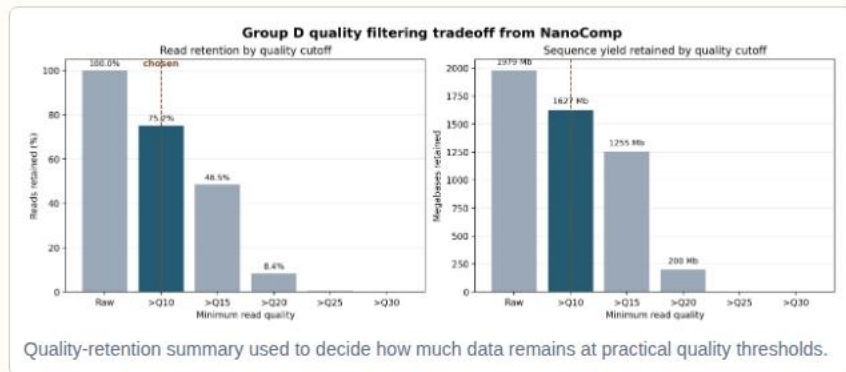
Same contig/overlap concepts as Lesson 9, but focused on one abundant bacterial species.

Reference comparison

dnadiff checks whether our bacterial assembly resembles a known reference genome.

QUALITY CONTROL

First question: are the reads usable?



Answer:

Yes. The reads are noisy, as Nanopore reads usually are, but still usable for long-read alignment and microbial screening.

Why these reports?

MinKNOW describes the sequencing run itself. NanoComp and NanoPlot summarize the reads after the fact, which is better for reproducible QC in the notebook. FastQC is common for short reads, but less informative for Nanopore length distributions.

Before interpreting biology, we looked at the run reports



These browser reports are the quick visual sanity check before we trust downstream analysis.

SUSPECT LINEUP

Six animals walked into the alignment step



House mouse

Good cameo, weak evidence.



Kangaroo

Memorable, but not our sample.



Domestic cat

Quietly sitting on the strongest signal.



Norway rat

Similar enough to test, not enough to win.



Golden jackal

Second place in some counts.



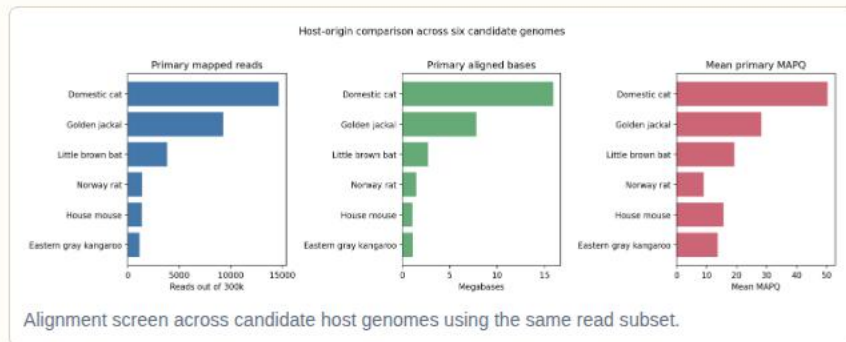
Little brown bat

Useful negative control energy.

The mouse got a slide. The cat got the evidence.

FIRST TRY

Let the reads vote: which genome do they align to?



Domestic cat had the strongest host signal.

It had the most primary mapped reads and the most aligned bases among the candidate genomes.

Why minimap2?

It is fast, widely used, and tuned for long noisy reads. BWA-MEM is excellent for many Illumina-style tasks, but minimap2 is the practical default for Nanopore host screening. BLAST would be too slow at this scale.

SECOND TRY

Not just more reads. Better reads.

Candidate	Primary mapped	Aligned bases	Mean MAPQ
1 Domestic cat	14,669	15.9 Mbp	50.3
2 Golden jackal	9,251	7.8 Mbp	28.3
3 Little brown bat	3,818	2.7 Mbp	19.2
Mouse / rat / kangaroo	1,155-1,393	1.1-1.5 Mbp	9.0-15.6

MAPQ is the confidence check.

A higher MAPQ means the aligner is more confident that a read belongs at that location.

Cat wins twice.

More mapped signal and much higher confidence than the nearest alternative.

HOST REVEAL

The sample origin is domestic cat

The candidate lineup was fun. The alignment scorecard made the call.

Top mapped reads

Top aligned bases

Top MAPQ



What "host locus" means in this project

Same coordinate idea as IGV.

In Lesson 5, we inspected a human coordinate in IGV. Here, after the host call, we inspect cat genome intervals instead.

"Functional" means annotated.

A GFF/GTF annotation file labels genome intervals as genes or other features, so a coordinate can become a named region.

1

Cat-aligned reads

Start only after the host call.

2

Cat annotation file

A table of gene coordinates from NCBI.

3

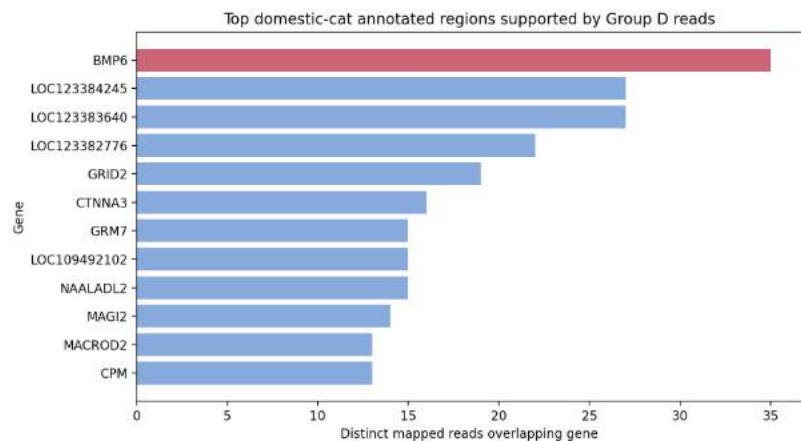
Overlap count

Which labeled region received the most read support?

So "host locus" just means a genomic interval inside the host reference. In our case, the host reference is domestic cat.

WHY BMP6?

BMP6 was our best annotated-region example



BMP6 had the highest read count among annotated cat gene overlaps in our summary.

Why it was chosen:

It was the top labeled gene region in the overlap table: 35 reads overlapped the BMP6 interval.

What BMP6 is:

BMP6 is a mammalian gene in a signaling family often discussed in development, bone/cartilage biology, and iron regulation.

What we are not saying:

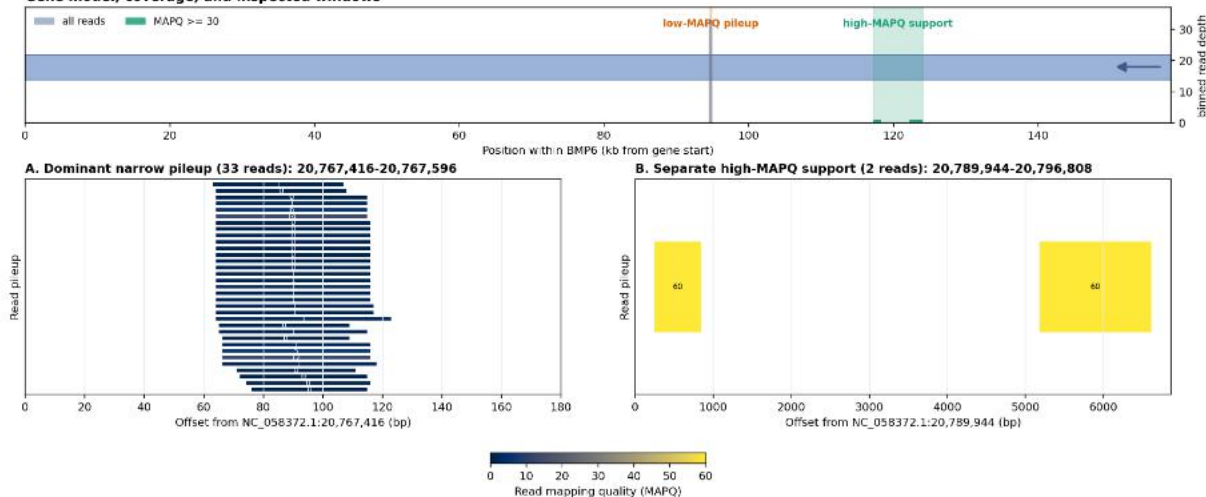
We are not saying this cat had a BMP6 mutation, phenotype, or special biology. We only found that this annotated region was represented in the host reads.

The IGV-style view is here because the class already used IGV

BMP6 host-region evidence: genome-browser-style view

NC_058372.1:20,672,770-20,831,131 | 158.4 kb gene | 35 overlapping reads | 2 reads with MAPQ ≥ 30

Gene model, coverage, and inspected windows



IGV-style BMP6 locus panel. It uses the same visual idea as the IGV homework: genomic coordinates, read pileup, and an annotation track.

Why an IGV-style panel?

Summary tables can hide awkward alignments. A genome-browser view lets us sanity-check whether the reads really sit near the annotated region.

Read it carefully:

The signal is concentrated and should be treated as supporting evidence, not as a functional discovery.

A fecal sample rarely travels alone

After host screening, the next question was: who else is in there?

For that, we used taxonomic classification rather than host-style whole-genome alignment.

Why Kraken2?

Kraken2 is fast and works well when the question is "what taxa are represented here?" Centrifuge is another lightweight classifier, MetaPhlAn is strong for marker-gene profiles, and KrakenUniq adds extra support information. Kraken2 was the best fit for a quick, reproducible species screen.

Plain version:

Host alignment asks "which animal genome?" Kraken2 asks "which microbial names do these reads most resemble?"

NEW METHOD FOR THIS PROJECT

Kraken2: using k-mers for taxonomy instead of assembly

1

Use a familiar idea

From assembly: split sequences into k-mers.

2

But ask a different question

Instead of building contigs, compare k-mers to a taxonomy database.

3

Report likely taxa

If enough k-mers point to a taxon, Kraken2 reports that label.

Why this fits the task:

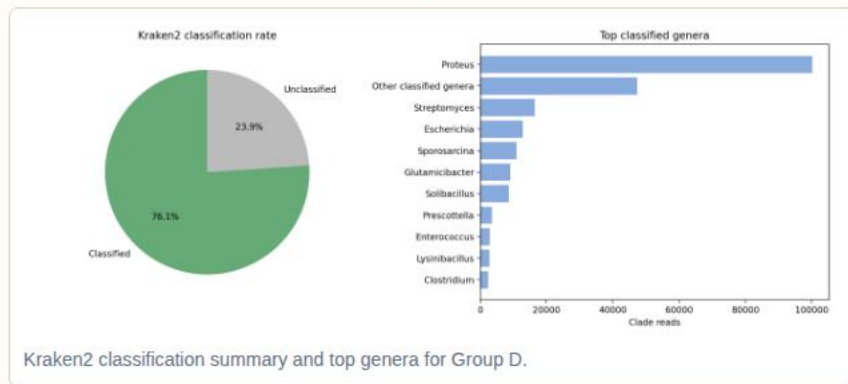
The assignment asks for microbiome composition, diversity, and top taxa. Classification is exactly that question.

Why we still keep caveats:

The answer depends on the database. Closely related organisms can share many k-mers.

MICROBIAL SCORECARD

Proteus dominates the genus-level profile



35.11%

Proteus at genus level

5.77%

Streptomyces

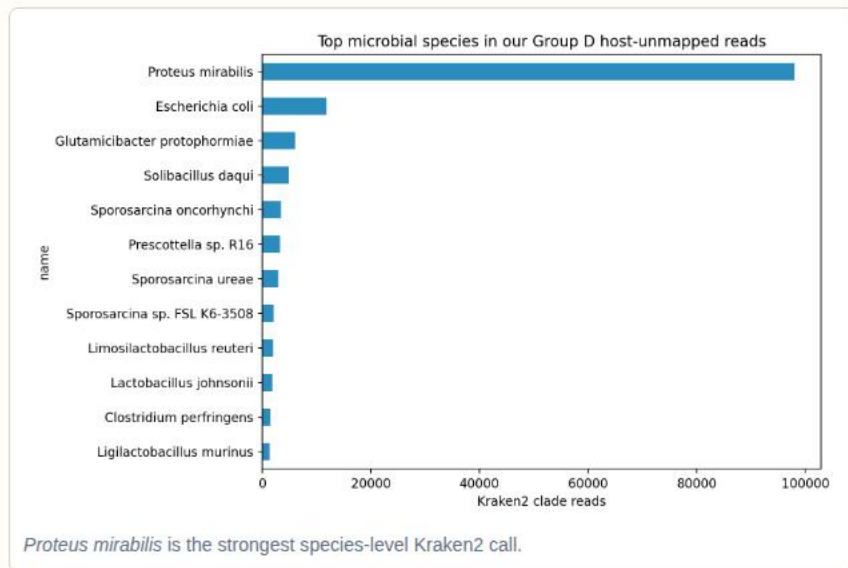
4.49%

Escherichia

3.84%

Sporosarcina

The dominant species is *Proteus mirabilis*



Top species call:

Proteus mirabilis, 34.35% of reads in the Kraken2 report.

The biological story so far:

A domestic cat fecal sample with a strong *Proteus mirabilis* microbial signal.

Lesson 9 idea, now on one abundant microbe

1

Pick the target

Use Kraken2 to choose the most abundant species-level signal.

2

Assemble its reads

Use Flye to turn long ONT reads into contigs.

3

Validate the result

Compare those contigs to a reference genome with dnadiff.

What Flye did:

Flye is the assembler: it reconstructs contigs from noisy long reads. It is not the classifier and not the validation step.

Why Flye?

Flye is built for noisy long reads and is commonly used for bacterial Nanopore assemblies. Canu is powerful but heavier, Raven is fast and lightweight, and Shasta is excellent in some long-read contexts. Flye was a balanced choice for this final-project dataset.

ASSEMBLY TARGET

Can we reconstruct the microbial co-star?

Why assemble *Proteus mirabilis*?

It had the strongest species-level signal, so it was the best candidate for a focused assembly attempt.

What we expected:

Because this is a mixed fecal sample, a fragmented assembly is normal. A perfect one-contig genome would be surprising.

3.86 Mbp

assembly size

24

contigs

577,987 bp

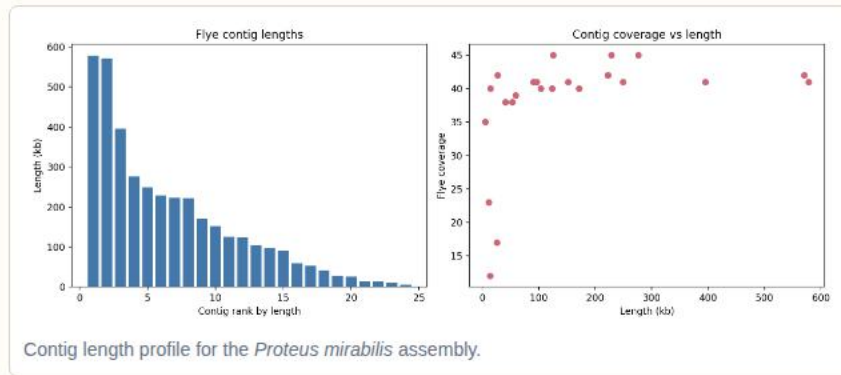
longest contig

249,377 bp

N50

ASSEMBLY RESULT

The assembly is fragmented, but biologically useful



What this means:

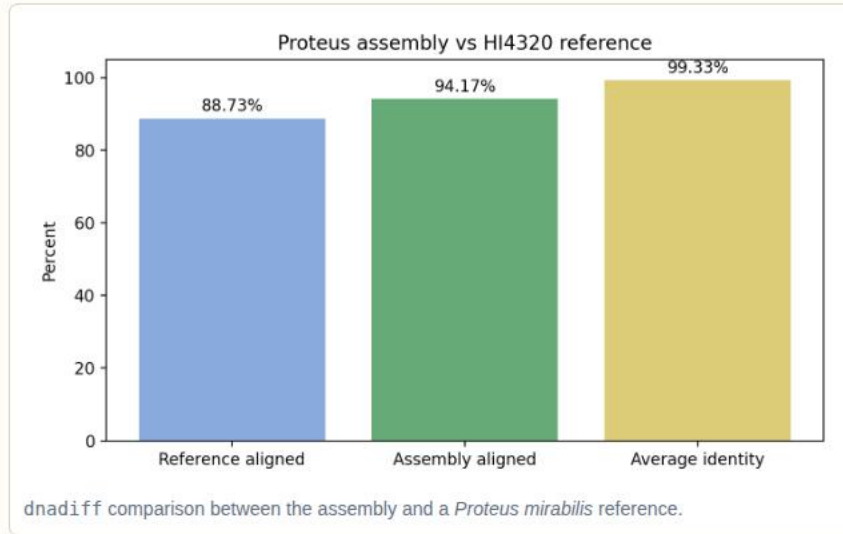
A perfect bacterial assembly would be closer to one complete chromosome. Here we got a partial but coherent assembly from a mixed fecal sample.

Good enough for:

Checking whether the dominant microbe can be reconstructed and compared against a reference.

VALIDATION

Does the assembly look like the reference?



Average identity: 99.33%

The assembled sequence aligns very closely to the reference where it aligns.

Coverage is not complete.

88.73% of the reference and 94.17% of the assembly aligned in the comparison.

Why dnadiff?

dnadiff is like a whole-genome diff tool. It aligns our assembly to a reference and reports how much matches and how similar the aligned parts are. QUAST is a strong alternative for broader assembly QC.

WHAT WE TRUST

Strong conclusions, careful caveats

We trust these results

- Host origin is domestic cat based on read count, aligned bases, and MAPQ.
- *Proteus mirabilis* is the dominant species-level microbial call.
- The Proteus assembly is close to a reference over aligned regions.

We are careful about these points

- BMP6 is a selected functional-region example from read overlap, not a phenotype claim.
- Kraken2 depends on the database and can misclassify reads from close relatives.
- The assembly is fragmented and from a mixed sample, so it is not a finished genome.

FINAL ANSWER

The short version

Host:

Domestic cat.

Functional region example:

BMP6 gene region, chosen because it had the strongest annotated host-read overlap.

Dominant microbe:

Proteus mirabilis.

Assembly:

3.86 Mbp, 24 contigs, N50 249,377 bp, 99.33% identity.

The cat did it.

Proteus mirabilis left the microbial fingerprints.

Full report

Curated GitHub Pages report

Notebook

Rendered reproducible analysis

PDF deck

Downloadable presentation